

Module 3: The Ionosphere

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1 Overview

Most of the sun's EUV and X-ray radiation incident on the Earth is absorbed in the upper atmosphere, and these photons have enough energy to ionize neutral molecules and atoms. The result is a "layer" of the atmosphere that is partially ionized, called the ionosphere. In this module we will discuss the origin of the ionosphere, its composition, and its different layers. We'll look at how it varies during the day, through seasons, and over the solar cycle, and a few of its more complex variations and unusual structures. The main effects of the ionosphere are primarily twofold: the ion and electron plasma causes spacecraft to become charged, leading to the potential for destructive discharging; and the ionosphere affects radio wave propagation, including both beneficial and deleterious effects. Finally, *Hidden Figures* and other classic space movies look at what an artificial ionosphere does to spacecraft communications during re-entry into the atmosphere.

2 Key Points

- The ionosphere is produced mostly by solar ultraviolet and X-ray radiation, with a small contribution from galactic cosmic rays and other charged-particle precipitations.
- The ionization density peaks during the day, and diminishes at night due to recombination
- The ionosphere is divided into D, E, and F-regions, where different wavelengths ionize different atmospheric constituents
- Ionosphere dynamics are controlled by fluid and plasma motions in the presence of the Earth's magnetic field and other forcing
- The ionosphere controls the propagation of radio waves below the plasma frequency
- The plasma environment leads to spacecraft charging, which can have deleterious effects on spacecraft electronics

Reading:

- Required: Pisacane, Chapter 13
- Optiona: Tascione, Chapters 7–9

3 Theory

The ionosphere is created by solar ionizing radiation. Consider a simple exponential atmosphere consisting of a single type of molecule, and a single type of ionizing radiation from the sun. What would the ionosphere look like? As the radiation reaches lower altitudes, there is more atmosphere to absorb it, and so ionization increases. However, as we descend further in altitude, the amount of ionizing radiation decreases because

it has been absorbed at higher altitudes. Thus, there must be a peak in the ionization at some altitude and a layer of ionization with some thickness and distribution in altitude.

This is the basic concept of a *Chapman Layer*¹, which we will now derive.

A constant average spectral flux of radiation, F_ν in $\text{W/m}^2/\text{nm}$, is incident upon the top of the atmosphere at an angle θ relative to the zenith. The atmosphere density varies with altitude as in the previous module:

$$\rho = \rho_0 e^{-z/H} \quad (1)$$

The spectral intensity I normally has units of $\text{W/m}^2/\text{nm}/\text{str}$. The “str” here stands for steradian, which is an angle in two dimensions. A sphere has 4π steradians just as a circle has 2π radians. A hemisphere has 2π steradians; just the top half of a sphere. In general, a cone with a half angle θ has $\Omega = 2\pi \int_0^\theta \sin \theta d\theta = 2\pi(1 - \cos \theta)$ steradians. Steradians are used to describe the angular distribution over which something is emitted or collected, or the two-dimensional field-of-view of a collecting instrument.

We describe the change in I using Beer’s Law, which tells us the intensity after traversing a path s :

$$I = I_0 \exp \left\{ - \int_0^s k \rho ds \right\} \quad (2)$$

The Flux F , which has units of $\text{W/m}^2/\text{nm}$, and the intensity I are related by integrating the latter over the solid angle:

$$\pi F = 2\pi \int_{-1}^1 I(\mu) \mu d\mu \quad (3)$$

where $\mu = \cos \theta$ is used to simplify the notation; think of it as the angle dependence (so that the integral runs from 0 to π in θ , and the 2π in front accounts for the integration in the azimuth angle).

Note that the path length segment ds in equation (2) is related to the vertical path length dz by

$$ds = \sec \theta dz = \frac{dz}{\mu} \quad (4)$$

Then equation (2) can be written as

$$I = I_0 \exp \left\{ - \int_0^s \frac{k \rho}{\mu} dz \right\} \quad (5)$$

We typically write $d\tau = k \rho dz$, where τ is the optical depth of the atmosphere; it combines thickness (z), density (ρ), and absorption coefficient (k) to describe how much total absorption occurs. Then equation (5) is written as

$$I = I_0 \exp \left\{ - \int_0^s \frac{1}{\mu} d\tau \right\} = I_0 e^{-\tau/\mu} \quad (6)$$

If we take the derivative with respect to τ , we have

$$\frac{dI}{d\tau} = -\frac{I}{\mu} \quad (7)$$

or, written in terms of the flux F ,

$$\frac{dF}{d\tau} = -\frac{F}{\mu} \quad (8)$$

If we now integrate this equation and plug in the appropriate terms, we have the equation for the flux F as a function of altitude:

$$F = F_\infty \exp \left(-k \rho_0 H \sec \theta e^{-z/H} \right) \quad (9)$$

where F_∞ is the incident flux at the top of the atmosphere. This function is an exponential raised to another exponential.

¹Derived by [Sydney Chapman](#), one of the great pioneers in space physics

Ionization in the Atmosphere: Now we want to know how this flux distribution produces ionization in the atmosphere over a range of altitudes. F above is the energy per unit area per unit time passing through a certain altitude; the energy absorbed per unit volume per unit time is then dF/ds . The rate of ionization (electron-ion pairs created per unit volume per unit time), or “Production” in the lecture slides, is then

$$q = \frac{1}{E_i} \frac{dF}{ds} = \rho(z) \frac{Fk}{E_i} \quad (10)$$

where E_i is the energy required to produce one electron-ion pair; this value is usually taken as $E_i = 35 \text{ eV}^2$ per ionization pair. Plugging in for F and ρ assuming an exponential profile for ρ , we find

$$q = \frac{\rho_0 k F_\infty}{E_i} \exp\left(-\frac{z}{H} - k_\nu \rho_0 H \sec \theta e^{-z/H}\right) \quad (11)$$

A typical ionization rate q is plotted in Figure 1. This profile shows the rate of ionization production, in pairs/m³/sec, from solar radiation, where a “pair” consists of one positive ion and one free electron produced in a single ionization event.

Electron Density: When we observe the ionosphere, we cannot measure the ionization rate directly; rather, we measure the electron density, which depends on the balance between ionization and recombination, and sometimes other processes (e.g., transport). We want to see how this ionization forms a steady-state electron density profile.

From the logic described earlier, there must be a peak in the ionization, $q_{\max}$, at some altitude $z_{\max}$. You can show that the peak ionization and altitude are given by

$$q_{\max} = \frac{F_\infty \cos \theta}{H} e^{-1} \quad \text{and} \quad \exp\left(\frac{z_m}{H}\right) = k \rho_0 H \sec \theta \quad (12)$$

We’ll next simplify the final solution with a new variable, z_1 , which is essentially the altitude relative to the peak altitude and scaled by the scale height:

$$z_1 = (z - z_{\max})/H \quad (13)$$

Then, the ionization profile, equation (11), can be re-written as

$$q = q_{\max} \exp[1 - z_1 - e^{-z_1}] \quad (14)$$

In a Chapman layer, the electron density n_e (electrons/cm³) is a balance between production, q , and loss, due to recombination of electrons and ions back into neutrals. This is the simplest statement of ion chemistry in the atmosphere. The electron density is then found from the solution to the following equation:

$$\frac{dn_e}{dt} = q - \alpha n_e n_i \quad (15)$$

$$= q - \alpha n_e^2 \quad (16)$$

where the first term on the right hand side is the production rate (q), and the second term is recombination, which is proportional to both the electron density n_e and ion density n_i (which is equal to n_e), and α is the recombination coefficient, which describes how quickly (or how easily) recombination occurs when an ion and an electron collide.

The solution to this equation has the form:

$$n_e(z, \theta) = \left(\frac{q_{\max}}{\alpha}\right)^{1/2} \exp\left\{\frac{1}{2} [1 - z_1 - e^{-z_1}]\right\} \quad (17)$$

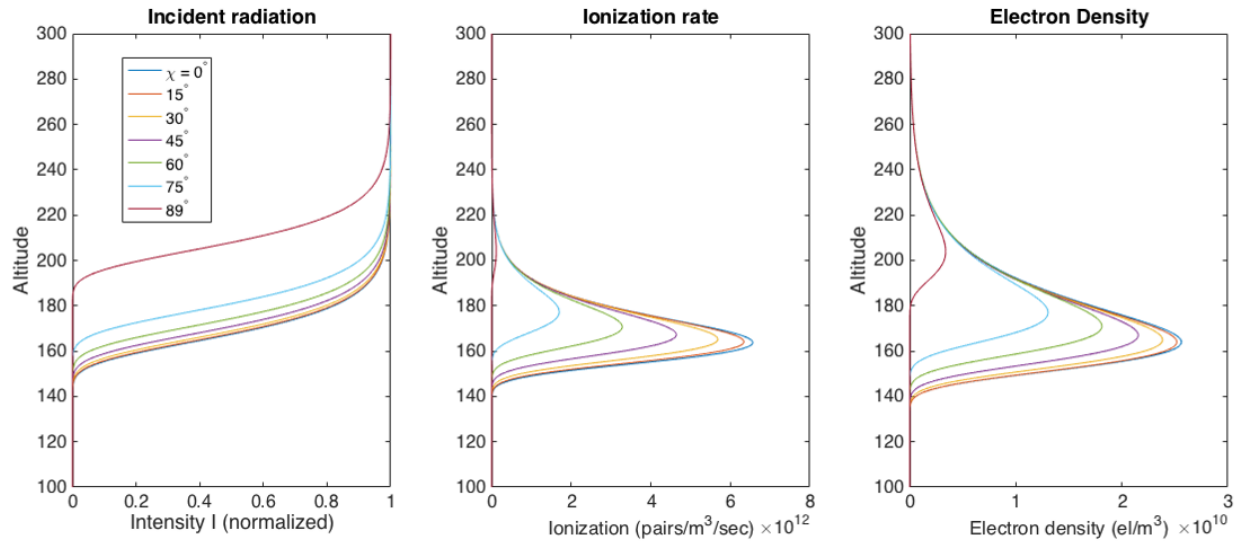


Figure 1: Plots of the incident radiation, ionization rate, and electron density Chapman profile.

This is the standard *Chapman profile*, which is a function of altitude and solar zenith angle θ ; it is plotted in Figure 1 for a few different angles. The Chapman profile provides a nice illustration of how any atmosphere forms a layer of ionization when exposed to solar ionizing radiation; as such, we expect all planets with atmospheres (Venus, Mars, Jupiter, Saturn... and probably most exoplanets) to also have ionospheres.

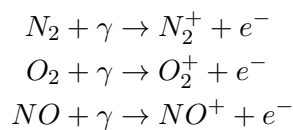
3.1 The Real Ionosphere

In reality, the density profile is more complex, due to variation in solar wavelengths, atmospheric composition profiles, and different absorption coefficients or cross sections (which are in turn a function of wavelength). Different wavelengths from the sun are absorbed by different molecules in the atmosphere (essentially they have different values of k , which is also a function of wavelength, $k(\lambda)$), and we have seen already that the atmosphere has different molecules or atoms at different altitudes. A “typical” ionosphere profile is shown in Figure 2.

3.1.1 The D-region

The D-region ranges from <70 km to about 90 km. The major constituents are NO^+ and O_2^+ . The D-region disappears (recombines) very rapidly, minutes after sundown. The major production and loss processes follow. We use γ to represent a photon.

Production:



²Remember an electron-volt (eV) is equal to 1.6×10^{-19} Joules; it is the energy imparted to an electron after accelerating through a 1 Volt potential difference.

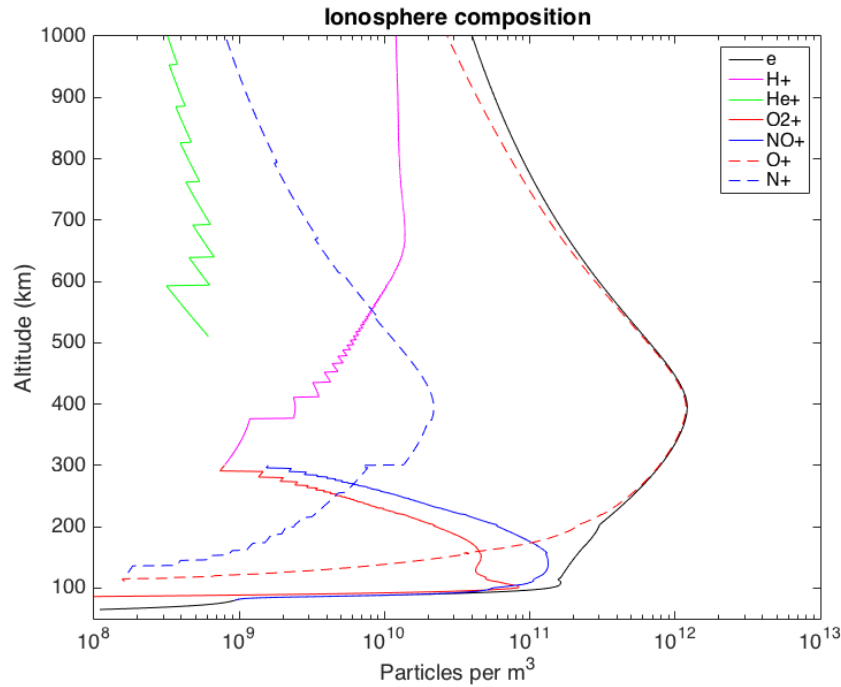
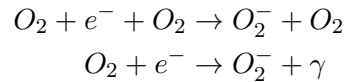
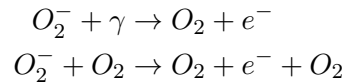


Figure 2: Ionosphere composition plot of altitude versus density.

Negative ions formed by attachment processes:



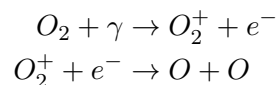
Detachment:



The D-region also contains heavy water cluster ions of the form $(H_2O)_nH^+$. We require more complex chemistry models to evaluate n_e profiles. The Glukhov-Pasko-Inan (GPI) 2009 model uses five equations (Figure 5) to determine the electron density, positive ion density, and negative ion density profiles. If we want any more information about the densities of specific ions, we need the more complex Sodankylä Ion and Neutral Chemistry (SIC) model, which includes hundreds of constituents and thousands of reactions (Figure 6).

3.1.2 The E-region

The E-region ranges from 90 km to 140 km. The major constituents are O_2^+ and NO^+ . The E-region recombines rapidly and often disappears before midnight, but commonly persists into the night. The major production and loss processes are simply:



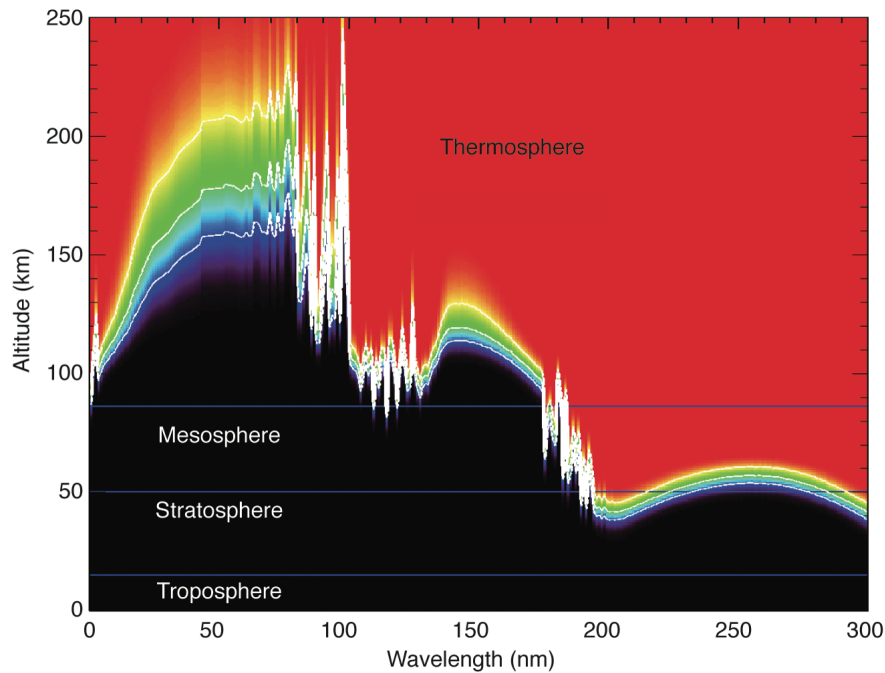
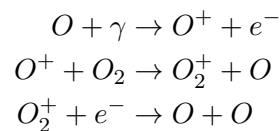


Figure 3: Layers of optical depth and energy absorption. As photons transit more and more neutral atmosphere, their absorption is total. The regions shown in black are unaffected by ionizing radiation from above. *top white curve* shows where $I(h)$ has decayed by $e^{-0.5}$; *middle white curve* by e^{-1} ; *bottom white curve* by $e^{-1.5}$; **Red areas:** $I(h)$ is basically I_∞ ; **Black areas:** $I(h)$ is basically zero.

3.1.3 The F-region

The F-region is split into the F1-region and the F2-region. The F1 region ranges from 140 km to 200 km. It is mostly made of O^+ and NO^+ . The F1-region mostly recombines after sundown, but pockets of ionization may remain. The F2-region from 200 km to 400 km is mostly O^+ , and it is persistent because of low collision rates and therefore low recombination, but the density still decreases after sundown. The F2-region is non-Chapman, because loss is not proportional to n_e^2 . The major production and loss processes in the F-region are:



3.1.4 The Topside Ionosphere

The topside ionosphere, at greater than 400 km, is mostly O^+ at lower altitudes and H^+ at higher altitudes, due to direct ionization of those dominant neutral species. It merges into the plasmasphere, which we discuss in Section 5.5.

4 Ionosphere Temperatures and Transport

Above ~ 110 km, ions, electrons, and neutrals have different temperatures. This simply means they have different velocity/energy distributions. Below 110 km, temperatures are made equal by collisions. Above,

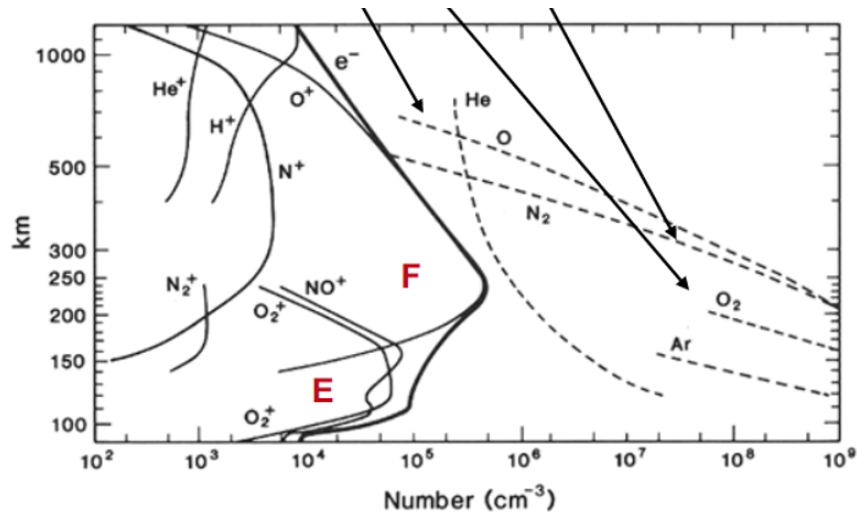


Figure 4: Altitude vs. density plot of multiple ions and neutrals with the E and F regions of the ionosphere labeled.

$$\begin{aligned}
 \frac{dN_e}{dt} &= Q - \beta N_e + \gamma N^- + \gamma_x N_x^- - (\alpha_d N^+ + \alpha_d^c N_x^+) N_e \\
 \frac{dN^-}{dt} &= \beta N_e - \gamma N^- - \alpha_i (N^+ + N_x^+) N^- - A N^- \\
 \frac{dN_x^-}{dt} &= -\gamma_x N_x^- - \alpha_i (N^+ + N_x^+) N_x^- + A N^- \\
 \frac{dN^+}{dt} &= Q - \alpha_d N_e N^+ - \alpha_i (N^- + N_x^-) N^+ - B N^+ \\
 \frac{dN_x^+}{dt} &= \alpha_d^c N_e N_x^+ - \alpha_i (N^- + N_x^-) N_x^+ + B N^+
 \end{aligned}$$

Figure 5: Equations used in the Glukhov Pasko Inan (GPI) 2009 D-region chemistry model.

collisions are rare, so each species gets its temperature through different heating processes (radiation absorption, convection, etc). Figure 7 shows the temperature profiles for ions, electrons, and neutrals.

A variety of processes move plasma in the ionosphere. Neutral winds drag ions along with them, if the collision frequency is high enough. Various forces cause plasma to “drift,” including electric and magnetic fields, gravity, and pressure.

5 Ionospheric Variations

This section provides just a brief description of some of the many ionospheric variations that are regularly observed. This is not nearly an exhaustive discussion of the changing ionosphere – entire textbooks cannot capture everything that we see. This is meant to provide just a glimpse of some of the most commonly observed variations.

5.1 Global variations

Total Electron Content The Total Electron Content (TEC) is the total number of electrons present along a path between a radio transmitter and receiver. It is calculated by integrating the electron density over a 1 m² column. Thus, TEC is measured in electrons per square meter. By convention, 1 TEC Unit or TECU

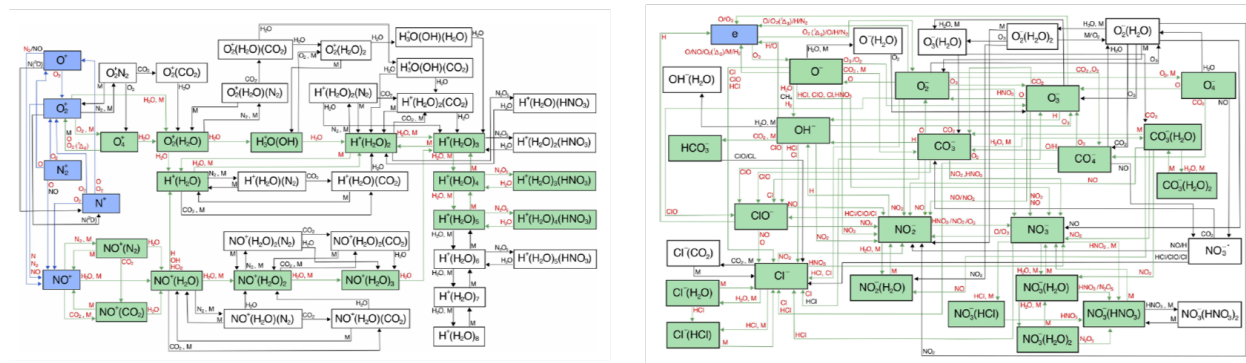


Figure 6: Schematic of the Sodankylä Ion and Neutral Chemistry (SIC) model, where each arrow represents an equation. This schematic does not include the neutrals also included in the model.

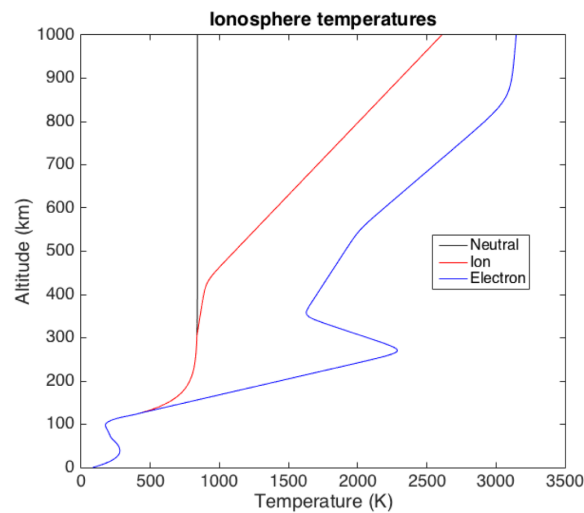


Figure 7: Example Temperature profiles of neutrals, ions, and electrons in the ionosphere, taken from the MSIS (for neutrals) and IRI models.

is 10^{16} electrons/m². Vertical TEC values in Earth's ionosphere can range from a few to several hundred TECU.

Daily and Solar Cycle variations We often characterize the ionosphere by its peak density, foF2, and the altitude where that occurs, hmF2. foF2 is a frequency in MHz, related to the electron density (#/m³) by the following equation:

$$f_c \simeq 8.979 \sqrt{N_{max}} \quad (18)$$

hmF2 is simply altitude in km. We can see a daily variation in foF2 and hmF2 in Figure 9.

Densities are much higher (order of magnitude) at solar maximum compared to solar minimum. This is because higher EUV/X-ray fluxes lead to higher ionization rates, and higher temperatures in the upper atmosphere at solar maximum lead to a higher density of neutrals available to be ionized. We can see the 11 year periodic variation in foF2 correlates to the solar cycle in Figure 10.

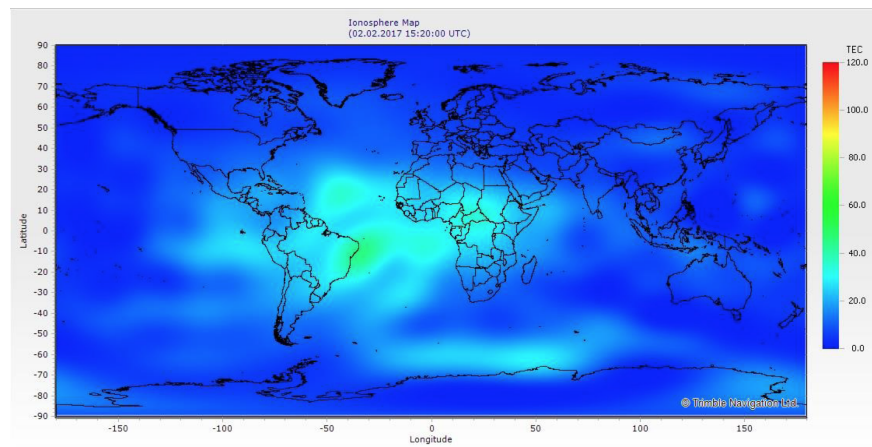


Figure 8: Total electron content (TEC) over the globe for 2017-02-02T15:20Z.

5.2 Equatorial variations

Equatorial Electrojet and Equatorial Anomaly Plasma physics involving the magnetic field lead to an intense current sheet that flows East in the dayside ionosphere called the equatorial electrojet, shown in Figure 11. It is restricted to a narrow region in latitude and about 110-130 km in altitude. Due to the electric current, an electric field arises. A plasma phenomenon called $E \times B$ drift creates a force that lifts the plasma in a vertical direction. The plasma rises, but above some altitude when the E-field no longer affects it, the plasma falls back down. Because plasma follows field lines, the plasma flow mimics the curve of a fountain, and this is known as the “Fountain Effect.” This is illustrated in Figure 12. This Fountain Effect leads to a dual band of enhanced plasma density on either side of the magnetic equator. We’ll talk about this more in the next module, because it requires understanding of how plasma moves in magnetic fields.

Equatorial Spread-F Plasma instabilities that occur right after sunset cause the F-region to take on an array of structures. Times scales for these phenomena range from seconds to hours, while spatial scales range from cm to tens of km. Figure 13 is a model of this Equatorial Spread-F.

Equatorial Plasma Bubbles Equatorial plasma bubbles are an instability due to heavier fluid on top of lighter fluid. This leads to rising bubbles with detailed structure and a wide range of spatial scales, with large density variation. In Figure 14, we can see Rayleigh-Taylor instabilities and Kelvin-Helmholtz instabilities.

5.3 Mid-Latitude variations

Sporadic-E At mid-latitudes, we regularly observe sporadic increases in E-region electron density, primarily at night, by orders of magnitude. The spatial scale of these patches of increased ionization is unknown. We have observed them through radio wave scattering. The effect is most likely due to shear winds driving metal ions from meteors into thin layers. This mid-latitude sporadic-E is illustrated in Figure 15.

5.4 High-Latitude variations

Polar Cap Absorption Solar Energetic Protons (SEPs) flow along open magnetic field lines and can access high latitudes and deposit their energy in the D-region of the ionosphere. This causes increased D-region density, and adds to absorption of radio waves discussed later in the Effects section. Figure 16 shows

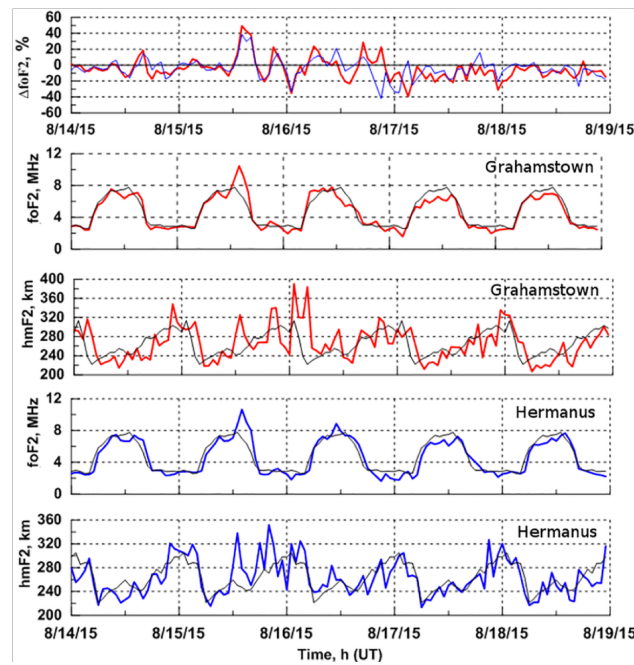


Figure 9: The second and third panels show foF2 and hmF2 versus time for Grahamstown, and the fourth and fifth for Hermanus. The top panel shows the difference in foF2 for the two locations. We can see the daily oscillation in foF2 and hmF2.

a schematic of SEP access into the polar regions on the left and solar protons detected by POES in the polar region on the right.

5.5 The Plasmasphere

The Earth's plasmasphere is a component of the inner magnetosphere located above the topside ionosphere. It is a region of dense, cold plasma that surrounds the Earth, and is trapped by the Earth's magnetic field. The boundary between the topside ionosphere and the plasmasphere is the location where thermal pressure and magnetic pressure are equal. In the plasmasphere, plasma becomes confined by the magnetic field. The structure of the plasmasphere is shown in Figure 17.

The plasmasphere varies with geomagnetic activity. One index of geomagnetic activity is the Planetary K-index (K_p), which is used to characterize the magnitude of geomagnetic storms. K_p quantifies disturbances in the horizontal component of earth's magnetic field with an integer in the range 0–9. A value of 1 corresponds to calm conditions, and a value ≥ 5 indicates a geomagnetic storm. $K_p = 9$ is very, very rare. The magnetic field measurements that go into the K_p calculation are made at various ground magnetometer stations; K_p is the average over all stations. K_p is quasi-logarithmic, so $K_p = 9$ is about 10 times stronger than $K_p = 3$. When K_p is high due to solar activity, the plasmasphere gets compressed. We call the outer boundary of the plasmasphere the *plasma*pause, which is the location where the density drops dramatically. Figure 18 models the plasma

6 Effects

6.1 Spacecraft charging

Spacecraft can lose electrons due to the photoelectric effect (energetic photons can knock out the electrons from spacecraft surfaces) and/or collect electrons as they fly through the ionosphere, plasmasphere, and

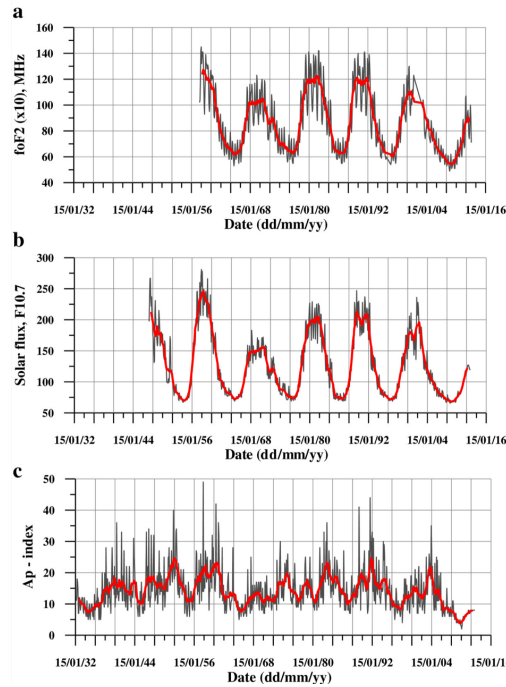


Figure 10: Plot a shows foF2 versus time, plot b shows F10.7 solar flux versus time, and plot c shows the Ap index (a geomagnetic activity index) versus time. We can see the how the variation of foF2 changes with the solar cycle.

energetic plasma regions (the plasmasheet and radiation belts) and thus become either negatively charged or positively charged. Charge from the ionosphere and/or plasmasphere can accumulate on spacecraft surfaces. The standard spacecraft charging equation describes the change in charge (Q) on a spacecraft surface due to various currents:

$$\frac{dQ}{dt} = I_e(V) + I_i(V) + I_{SE E_i}(V) + I_b(V) + I_{ph}(V) \quad (19)$$

where Q is the total charge on the surface, V is the voltage of the surface, I_e is the electron current from the plasma, I_i is the ion current from the plasma, $I_{SE E_i}$ is the secondary electron emission due to ion current, I_b is the backscattered electron current, and I_{ph} is the photoelectron current.

Let's consider an unbiased, conducting object in LEO. In LEO, the ion thermal velocity is less than the orbital velocity of the object, which is less than the electron thermal velocity. Thus, ions only impact on the ram side of the object. The ion current on the object is:

$$I_i = q_e n_i v_0 A_i \quad (20)$$

where q_e is the elementary charge, n_i is the ion number density, v_0 is the orbital speed and A_i is the area of the spacecraft area that collects ions (ram side). The electrons can impact the entire spacecraft, and the current is given by:

$$I_e = -\frac{1}{4} q_e n_e v_{e,th} A_e \left(e^{q_e V / k_B T_e} \right) \quad \text{for } V < 0 \quad (21)$$

$$I_e = -\frac{1}{4} q_e n_e v_{e,th} A_e \left(1 + \frac{q_e V}{k_B T_e} \right) \quad \text{for } V > 0 \quad (22)$$

where A_e is the area that collects electrons (the entire surface). Note that the electron current flowing *towards* the spacecraft surface is *negative*. The spacecraft will accumulate negative charge until the *negative*

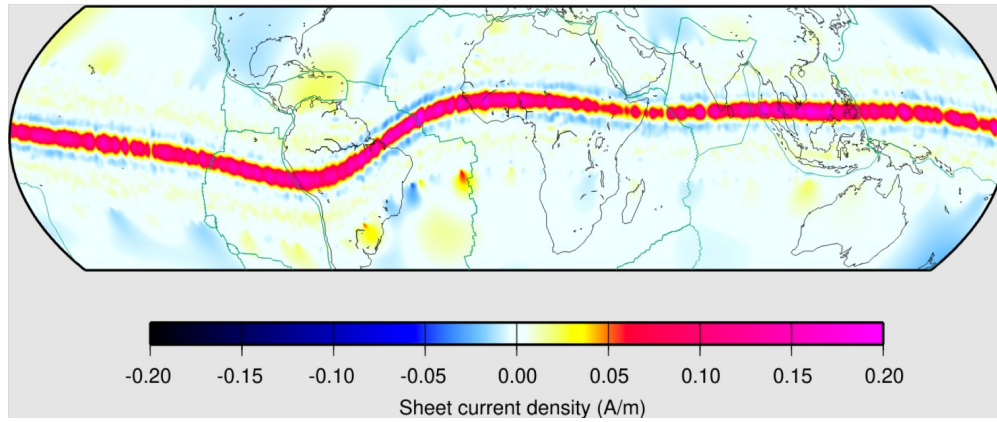


Figure 11: Equatorial electrojet.

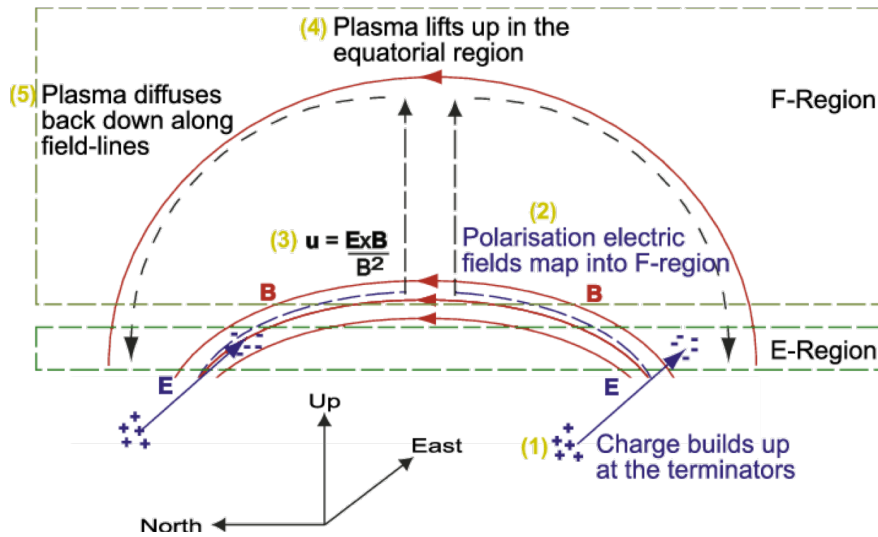


Figure 12: Equatorial anomaly fountain effect.

potential is enough to repel the excess electrons and balance the currents. In this state, the spacecraft is charged to the floating potential:

$$V_{fl} = \frac{kT_e}{q_e} \ln\left(\frac{4v_0 A_i}{v_{e,th} A_e}\right). \quad (23)$$

This is a closed-form equation, and we can simply plug in values for area, temperature, and thermal velocity to solve for the expected spacecraft potential.

In higher orbits, like GEO, spacecraft charging concerns change because the plasma temperatures increase dramatically, so that the ion thermal velocity far exceeds the spacecraft velocity. So instead of ion current being controlled by the spacecraft's orbital velocity, it is controlled by the plasma characteristics. In this case, the ion current is given by:

$$I_i = \frac{1}{4} q_e n_0 v_{i,th} A_i \left(e^{-q_e V / k_B T_i} \right) \quad \text{for } V > 0 \quad (24)$$

$$I_i = \frac{1}{4} q_e n_0 v_{i,th} A_i \left(1 - \frac{q_e V}{k_B T_i} \right) \quad \text{for } V < 0 \quad (25)$$

Assuming $n_i = n_e$, $A_i = A_e$, and $T_i = T_e$, and assuming $V < 0$ for now, the combination of the balance of

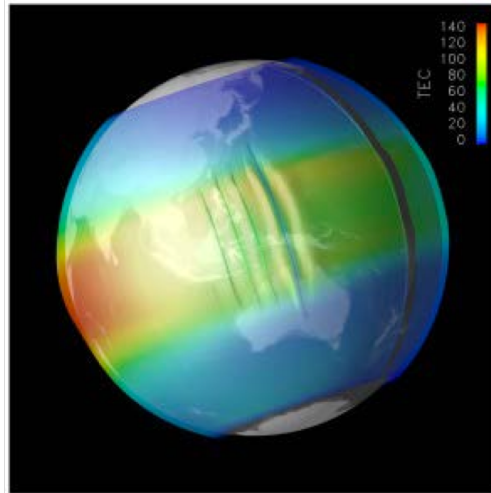


Figure 13: Structures forming in the F-region equatorial ionosphere just after sunset; these are measured as Equatorial Spread-F in radar observations.

the ion and electron currents leads to the following equation for the floating potential:

$$V_{fl} = \frac{k_B T_e}{q_e} \ln \left[\sqrt{\frac{m_e}{m_i}} \left(1 - \frac{q_e V}{k_B T_i} \right) \right] \quad (26)$$

Unfortunately, with the V on the right hand side, this is not a closed form solution, and one does not exist. However a rough approximation of the solution is that the floating potential is

$$V_{fl} \simeq -2.5 T_e \quad (27)$$

where T_e is the electron temperature in eV. At GEO orbit, the electron temperature could be as high as 10^7 K or nearly 1000 eV; in which case $V_{fl} = -2500$ V. Clearly at GEO spacecraft charging is a much more significant concern!

An important factor we have left out is the photocurrent I_{ph} . The photocurrent arises from solar energetic photons removing electrons from the spacecraft³; as electrons are removed from the spacecraft, the photocurrent leads to *positive* voltages on the spacecraft surfaces. The photocurrent from a positively-charged surface is given by:

$$I_{ph}(V) = \left(40 \frac{\mu\text{A}}{\text{m}^2} \right) A \cos \theta e^{-qV/k_B T_{ph}} \quad (28)$$

where $A \cos \theta$ is the surface area exposed to the sun, and $k_B T_{ph} \simeq 2$ eV is the average energy of a solar photon. The photocurrent is *positive*, because it is a flow of electrons *leaving* the spacecraft surface.

The photocurrent can be strong enough to dominate over the electron and ion currents on surfaces exposed to the sun, leading to large positive potentials.

When concerned about charging effects, we rarely care about the absolute potential; it is simply the potential of the spacecraft relative to a reference “ground”, the plasma some distance from the spacecraft. More significant effects occur when there is significant potential difference (i.e. voltage) between two regions of spacecraft; thus our major concern is *differential spacecraft charging*. Electrostatic Discharge is the transfer of charge through spacecraft material due to an electric potential (voltage) difference. It starts at -150 to -500 V and produces currents, interference, and electromagnetic pulses. In GEO, charging events most commonly occur in early morning. In LEO, discharge events occur over South America and at high latitudes (>70 degrees) where charging due to energetic particles is significant.

³due to the photoelectric effect, which of course is what Albert Einstein got his Nobel Prize for.

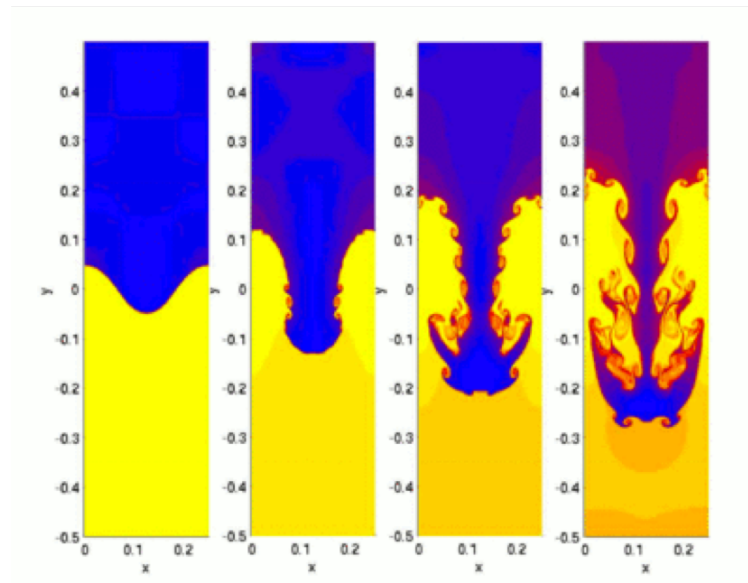


Figure 14: The Rayleigh-Taylor instability which occurs when a heavy fluid sits on top of a lighter fluid. As the instability grows from left to right, we also see evidence of the Kelvin-Helmholtz instability forming along the boundary of the two fluids.

6.2 Radio Wave Propagation

The ionosphere reflects and modifies radio waves used for communication and navigation. To discuss ionospheric effects on radio waves, let's first review some useful EM wave and plasma physics basics.

6.2.1 Electromagnetic Waves and Plasma Physics Review

A plasma is a gaseous mixture of electrons, ions, and neutral particles. The ionosphere is a weakly ionized plasma. An electromagnetic pulse or a radio wave can displace electrons from ions in a plasma. The resulting separation of charge sets up an electric field which attempts to restore equilibrium. Due to their momentum, the electrons will overshoot the equilibrium point, and accelerate back. This sets up an oscillation. The frequency of this oscillation is called the plasma frequency:

$$\omega_p = \sqrt{\frac{n_e q_e^2}{m_e \epsilon_0}} \quad [\text{rad/s}]$$

$$f_p = \frac{\omega_p}{2\pi} \quad [\text{Hz}]$$

where n_e is the number density of electrons, q_e is the electric charge, m_e is the mass of the electron, and ϵ_0 is the permittivity of free space. We typically use f for frequency when we are talking about radio waves, and ν (greek letter “nu”) for collision frequencies.

Recall from Module 1 that a radio wave consists of oscillating electric and magnetic fields. Here are some useful terms to know when discussing how EM waves interact with matter:

- *Full propagation*: little direction change; little energy loss
- *Reflection*: wave rebounds from a ‘surface’; direction change
- *Refraction*: wave bends through a different medium; direction change

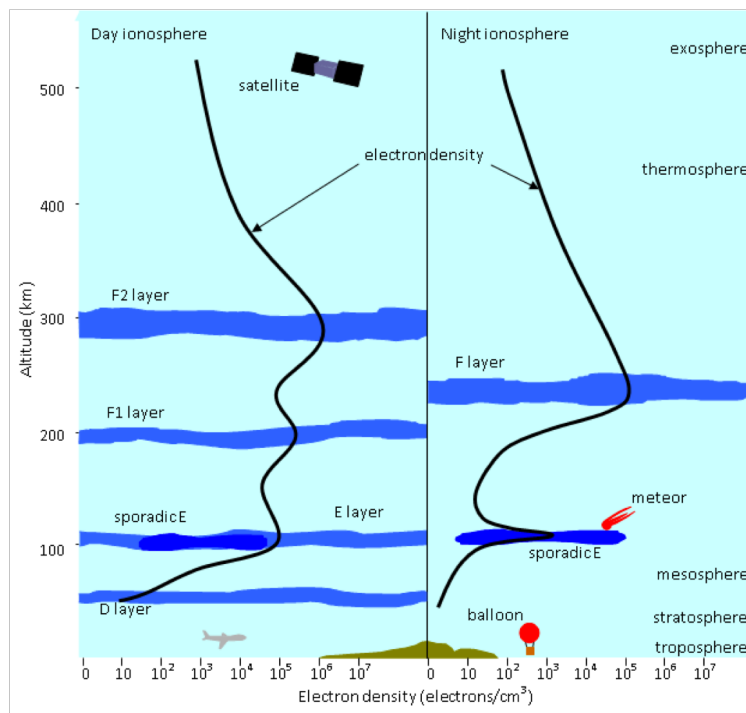


Figure 15: A basic illustration of Sporadic-E. It may occur in the daytime, but is more obvious at night since the background E-region is much weaker at night.

- *Scattering*: wave interacts with very small objects; direction change
- *Attenuation*: wave interacts with object and loses energy
- *Diffraction*: wave bends around a large object
- *Scintillation*: modulation of wave due to diffraction (sometimes all of the above; can be severe effect leading to loss of signal lock)

As a radio wave propagates from one medium to another, its propagation direction is refracted, based on the indices of refraction, n_1 and n_2 . Recall Snell's law:

$$\frac{\sin \theta_2}{\sin \theta_1} = \frac{v_2}{v_1} = \frac{n_1}{n_2} \quad (29)$$

where θ_1 and θ_2 are the angles of incidence and refraction and v_1 and v_2 are the phase velocities in the two media (see Figure 19).

6.2.2 Reflection of Radio Waves in the Ionosphere

The behavior of a radio wave in the ionosphere depends on the plasma frequency. Plasma frequency ω_p , defined above, is directly related to electron density, with ω_p is proportional to $\sqrt{n_e}$. There are four major options for what happens to a radio wave in the ionosphere:

- Radio waves below ω_p are *reflected*. The electrons are “shaken” and re-radiate. More specifically, the local charged particles have sufficient time to rearrange themselves so as to cancel out the oscillating electric field and thereby screen the rest of the plasma from the oscillating E-field.

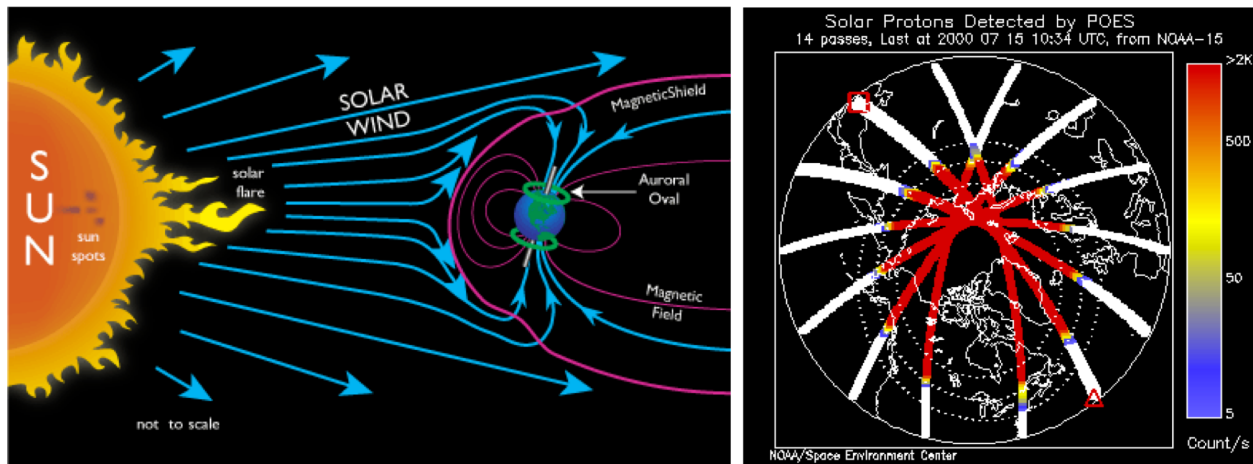


Figure 16: SEPs accessing the polar region (left) and solar protons detected by POES in the polar region (right).

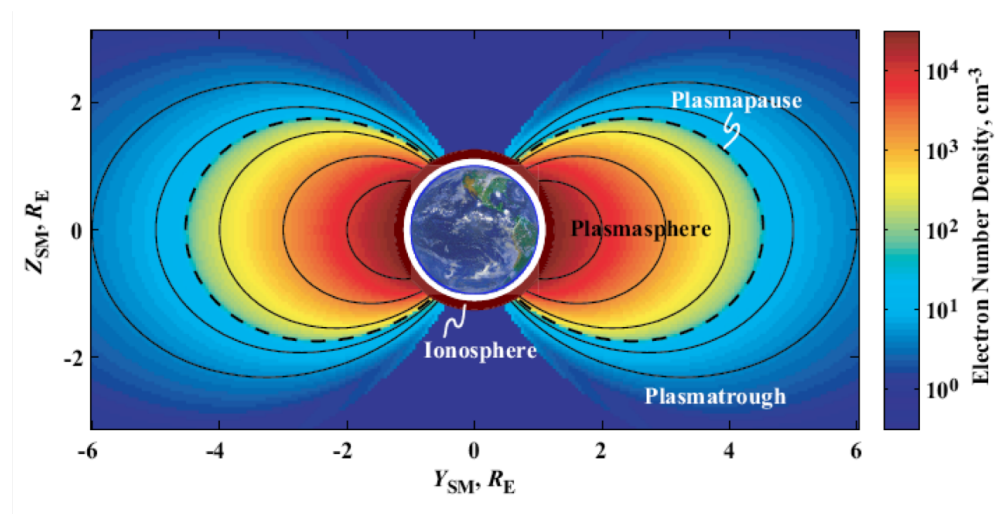


Figure 17: Plasmasphere model of electron density.

- Radio waves above ω_p *pass through* the ionosphere. Electrons cannot respond fast enough to reflect the wave. In other words, the particles do not have time to adjust themselves to produce the screening effect from above, and the wave passes through.
- The radio wave is *absorbed* in the ionosphere due to collisions between the oscillating electrons and the neutral molecules.
- The wave is *scattered* in random directions by plasma irregularities.

Let's focus on the implications of the first two options. The first point means that we can communicate over-the-horizon with frequencies near or below ω_p . If you hear the phrase "Earth-ionosphere waveguide," it is referring to this over-the-horizon communication phenomena.

The second point means we must use frequencies above ω_p to talk to satellites. Satellite communications tend to use high frequency signals: Ultra High Frequency (UHF), 300 MHz - 3 GHz and Super High Frequency (SHF), 3 - 30 GHz. The critical frequency of the ionosphere (foF2 from Section 5.1) represents the minimum radio frequency capable of passing completely through the ionosphere.

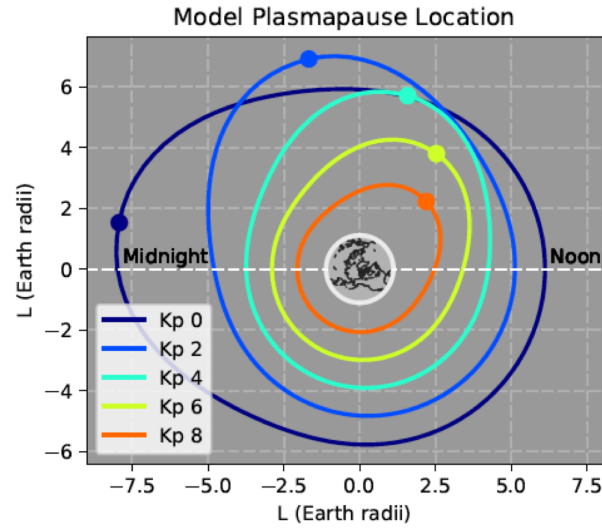


Figure 18: Plasmopause location at different Kp conditions.

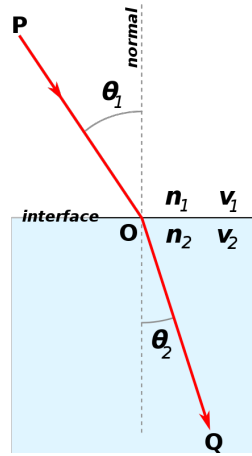


Figure 19: Snell's Law.

Mathematically, the behavior of a radio wave in a plasma can be viewed in terms of an index of refraction. The (simplified) of refraction for a radio wave in a plasma is given by⁴:

$$n = \sqrt{1 - \frac{\omega_p^2}{\omega^2}} = \sqrt{1 - \frac{8.1 \times 10^{-5} n_e}{f^2}} \quad (30)$$

where n_e is the electron density in cm^{-3} and f is the wave frequency in MHz. For a given n_e , a high-frequency wave will suffer less refraction than a lower frequency wave, since n is closer to 1 for a high-frequency wave. The above expression for refractive index becomes imaginary for certain values of n_e/f , implying reflection of the radio wave. For a vertically-propagating wave, this occurs at the critical value (where $n = 0$):

$$f_c = 9 \times 10^{-3} \sqrt{n_{ec}} \quad (31)$$

⁴The full equation for the index of refraction is known as the Appleton-Hartree equation, and becomes much more complicated as it includes the effect of the Earth's magnetic field and collisions between electrons and atmospheric neutrals.

This is the same as the foF2 equation from Section 5.1, with new unit of n_e ! Again, this is the the lowest frequency radio wave capable of penetrating the ionosphere.

As a wave with frequency lower than the critical frequency propagates, it refracts as it propagates through the progressively increasing electron density. To trace it's path, you need to solve Snell's law repeatedly in thin layers of the ionosphere, as illustrated in Figure 20. Thus, we are treating the ionosphere as successive layers, and look at refraction from one layer to the next. The end result is that the ray “bends” and turns back to the ground, rather than reflecting from a sharp boundary.

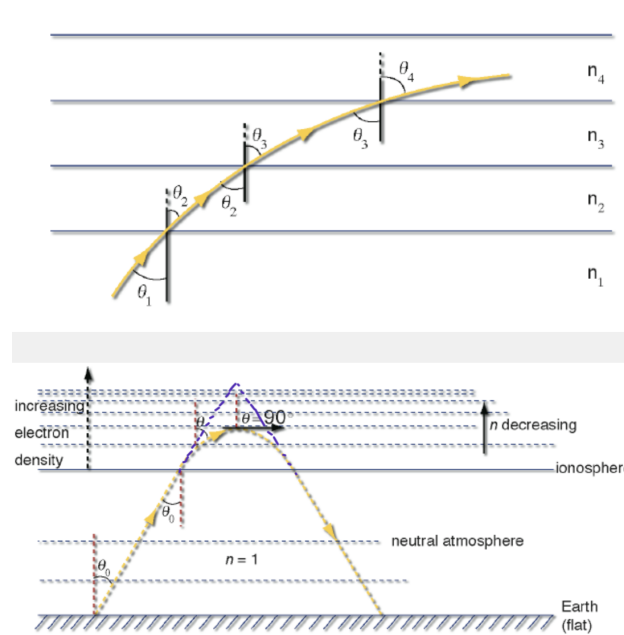


Figure 20: Snell's Law in the ionosphere.

In the F-region, f_c is $\sim 3\text{--}30$ MHz, so higher frequencies pass through the ionosphere, with some refraction over-the-horizon radio. In the E-region, f_c is $\sim 1\text{--}2$ MHz, but sporadic-E (Section 5.3) increases f_c up to 100 MHz. The D-region is more complex, and will be discussed below. While the application of Snell's law and the critical frequencies describe a good portion of the behavior of radio waves in the ionosphere, ionospheric variability from space weather makes the ionospheric effects on propagating signals are highly variable.

6.2.3 Radio Wave Absorption in the Ionosphere

In the D-region, our model breaks. Lots of neutrals means a higher collision frequency. Our index of refraction becomes more complicated. We get absorption of MHz waves and reflection of waves below ~ 100 kHz. This reflection is why VLF waves (below ~ 50 kHz) are used for long-range communications with submarines. Absorption of the D region occurs when electrons are excited by waves with $f < f_c$, and they collide with neutrals. Some of the EM wave energy gets transferred to heat via enhanced collisions between electrons/ions and neutrals, and the radio waves suffer significant absorption. Figure 21 shows absorption of a 30 MHz radio wave due to electron precipitation from the radiation belts. A riometer is a passive instrument that measures D-region absorption by monitoring cosmic noise at ~ 30 MHz.

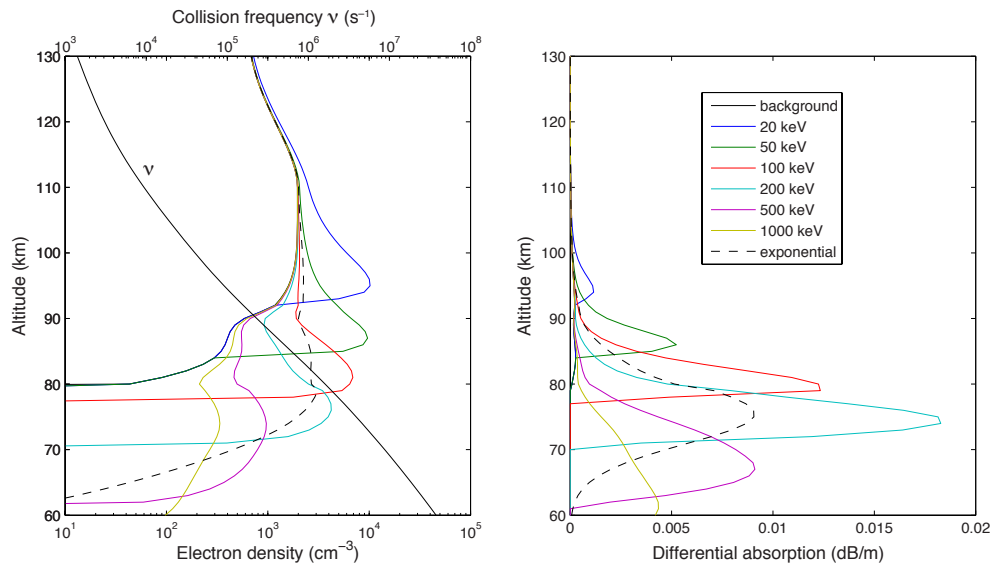


Figure 21: Absorption of a 30 MHz radio wave in the D-region ionosphere. The left plot shows electron density profiles created by electron precipitation of different energies from the radiation belts; the right plot shows the resulting absorption of 30 MHz radio waves.

6.2.4 Solar Flares and the Ionosphere

Recall that solar flares are large eruptions of electromagnetic radiation from the Sun lasting from minutes to hours. The sudden outburst of electromagnetic energy travels at the speed of light, therefore any effect upon the sunlit side of Earth's exposed outer atmosphere occurs at the same time the event is observed.

A Sudden Ionospheric Disturbance (SID) is ionospheric response to a solar flare (X-rays). The burst of X-rays ionize the D-region, causing a huge increase in D-region electron density by orders of magnitude. The higher n_e leads to higher radio wave absorption. HF radio signals can become degraded or completely absorbed. This results in a radio blackout—the absence of HF communication, primarily impacting the 3 to 30 MHz band. This is why NOAA SWPC classifies flares on a “Radio Blackouts” scale, which we discussed in the Sun module. In addition, the resulting lower D-region reflection height perturbs VLF signals.

Also, solar energetic particles, most protons of MeV to 100s of MeV, associated with intense Solar Flares and CME have direct access to the polar cap region (where Earth's magnetic field lines are connected to the solar wind) and reach at low altitude, leading to large increases in D-region ionization, and thus producing large increases in ionospheric attenuation of high-high-frequency radio waves, which is termed polar cap absorption (PCA).

When using over-the-horizon radar, there is a maximum frequency we can use, above which waves pass through the F-region. There is also a minimum frequency that we can use because $dA/dl = 4.6 \times 10^{-5} n_e \nu / \omega^2$; lower frequencies are subject to higher absorption due to the dependence $1/\omega^2$ (Why?). Therefore, we have a usable “Frequency Window”; after a major X-ray flare, absorption can increase to prevent any useful communication. This is called Short-Wave Fadeout, a particular type of SID that can severely hamper HF radio users by causing increased ionization and signal absorption, which may last up to hours.

6.2.5 GPS and TEC

The U.S. Global Positioning System (GPS), European Galileo, Russian GLONASS, and Chinese BeiDou (BDS) are global satellite navigation systems that transmit navigation signals at multiple frequencies and are

affected by propagation through the ionosphere. In the absence of space weather, GPS systems compensate for the “average” or “quiet” ionosphere, using a model to calculate its effect on the accuracy of the positioning information. But when the ionosphere is disturbed by a space weather event, the models are no longer accurate and the receivers are unable to calculate an accurate position based on the satellites overhead.

Geomagnetic storms create large disturbances in the ionosphere. The currents and energy introduced by a geomagnetic storm enhance the ionosphere and increase the total height-integrated number of ionospheric electrons, or the Total Electron Count (TEC, Section 5.1). GPS systems cannot correctly model this dynamic enhancement and errors are introduced into the position calculations. This usually occurs at high latitudes, though major storms can produce large TEC enhancements at mid-latitudes as well.

7 Missions

7.1 ICON

[ICON, the Ionospheric Connection Explorer](#), was designed to understand drivers of ionospheric variability, explain how energy/momentum from lower atmosphere reach the space environment (e.g. gravity waves), and explain how drivers create extreme conditions observed during solar-driven geomagnetic storms. There are four main instruments. *MIGHTI* is a Michelson Interferometer to measure winds and temperatures. *FUV* is a (creatively named) FUV imager that observes UV emissions of N₂ and O to determine O/N₂ ratio. *EUV* images 83.4 nm emission from O, and resonantly scattered by O⁺: gives ion density. *IVM* is the ion velocity meter, which uses a Retarded Potential Analyzer (RPA) to measure relative velocity of ions, therefore winds, as well as temperature and density.

7.2 AMPERE

[AMPERE, the Active Magnetosphere and Planetary Electrodynamics Response Experiment](#), uses measurements of magnetic fields to determine the currents that created them.

7.3 COSMIC and GPS

We can measure the TEC in the ionosphere using radio occultation and path delay. Then, we essentially take a ton of TEC measurements and throw them into a giant inversion model to get 4D ionosphere. This is done with both GPS (Global Positioning System) to ground signals and spacecraft to spacecraft signals with [COSMIC, the Constellation Observing System for Meteorology, Ionosphere and Climate](#) (a GPS receiver instrument on Taiwanese FORMOSAT-3 spacecraft).

8 Movie: Hidden Figures

Synopsis from Fandango:

Hidden Figures is the incredible untold story of Katherine Johnson (Taraji P. Henson), Dorothy Vaughan (Octavia Spencer) and Mary Jackson (Janelle Monáe)-brilliant African-American women working at NASA, who served as the brains behind one of the greatest operations in history: the launch of astronaut John Glenn into orbit, a stunning achievement that restored the nation's confidence, turned around the Space Race, and galvanized the world. The visionary trio crossed all gender and race lines to inspire generations to dream big.



Figure 22: Hidden Figures (2016).

A serious space movie that doesn't end with everyone dying in the sun or involve immature 1980s jokes about the Schwartz. In my opinion this is a fantastic movie for a whole variety of reasons (and I hope John Glenn really had the personality that he is portrayed with). But this movie actually never mentions the ionosphere. I have yet to find a decent space movie that mentions the ionosphere at all; perhaps spacecraft charging and radio wave reflection are not cinematically appealing?

Instead, we'll use this movie to probe a classic (and still relevant) re-entry phenomenon. When a spacecraft re-enters the atmosphere, there's a period of time where communications are lost. Why does this happen? The clip to pay attention to is posted on Canvas.

Problems

1. **(20) Plotting the Ionosphere:** Use the International Reference Ionosphere (IRI-2012) hosted by the CCMC Run the model to create an output file changing only the following parameters:

Variable	Value
Year, Month, Day, Time	2006, Dec. 21
Time	Local
Hour of day	Noon
Latitude, Longitude	40 N, -105 E (255 W)
Altitude range:	0 to 1000 km
Step size	5 km
IRI Model Parameters	e^- , O^+ , H^+ , O_2^+ , NO^+ and He^+

DO NOT submit the figure that is generated on the website; download the data (“View Raw Output” at the top; copy and paste into a text file) and then make your own plots.

- i) Plot altitude (on the y-axis) vs. number density (log scale, on the x-axis) for electrons, O^+ , H^+ , O_2^+ , NO^+ , and He^+ . Use your plot to answer the following questions:
 - ii) Below 500 km, over what altitude range does NO^+ dominate? At what range of altitudes does O_2^+ contribute more than 20% of the total?
 - iii) What is the total electron content (TEC)? What is the TEC in the topside ionosphere (region above the peak electron density)? What is the TEC in the bottom side ionosphere (region below the peak electron density)? Report your answers in TEC units (TECU).
2. **(10) Peak Ionization:** In a Chapman layer, the peak ionization production rate, $q_{\max}$, is related to the electron density $n_{e,\max}$ and the effective recombination coefficient α_{eff} by $n_{e,\max} = [q_{\max}/\alpha_{\text{eff}}]^{1/2}$.
- i) Using the following parameters, determine $q_{\max}$ for each layer of the ionosphere:
- | Region | Peak Altitude | $n_{e,\max}$ (cm^{-3}) | α_{eff} (cm^3/s) |
|--------|---------------|-----------------------------------|--|
| E | 110 km | 1.5×10^5 | 10^{-8} |
| F1 | 200 km | 2.5×10^5 | 7×10^{-9} |
| F2 | 300 km | 5.0×10^5 | 5×10^{-9} |
- ii) Explain why, when all other factors are kept constant, the ionization peak altitude ($z_{\max}$) of a layer moves upward with increasing solar zenith angle χ .
3. **(10) Spacecraft Charging in LEO:** What is the floating potential of an unbiased 6U CubeSat in low-Earth orbit at 400 km altitude, flying with its 20×30 cm face in the ram direction? Assume an electron temperature of 1000 K. What assumptions did you have to make?
4. **(10) Spacecraft Charging in GEO:** Consider a spacecraft in GEO with one surface in sunlight (normal to the sun’s direction) and one surface in shadow. Use equation (19) and the relevant equations that follow to calculate the floating potential on each of these two surfaces. Assume a plasma density of $n_e = n_i = 0.5 \text{ cm}^{-3}$, and a plasma temperature $T_e = T_i = 5 \times 10^6$ K. You can ignore the SEE and backscatter currents. What is the potential difference between these two surfaces?
- [Hint: you can assume the potential is negative in shadow and positive in sunlight. Note that you’ll probably need to solve the equation for dQ/dt numerically.]
5. **(10) Radio Wave Propagation:** Using the IRI profile you created in Problem 1, i) what is the minimum radio frequency that can be used to communicate from ground to space? ii) From which ionospheric region does this frequency reflect? iii) From what altitude does a 2 MHz radio wave reflect?
6. **(10) Radio Blackout:** During spacecraft re-entry, as demonstrated in Hidden Figures, the spacecraft temporarily loses the ability to communicate with the ground. i) Describe in a few sentences why this happens. ii) If the module normally communicates with the ground at 2282.5 MHz, what must the minimum electron density be in the plasma sheath region?
7. **(10) GPS Delays:** i) Calculate the path length delay through the ionosphere for 10 TECU, at $L1 = 1575.42$ MHz and at $L2 = 1227.60$ MHz. ii) Explain how you can use dual-frequency GPS to reduce the path length (and thus position) errors due to the ionosphere.

8. **(10) D-region Signal Attenuation:** Calculate the attenuation (in dB) of a 10 MHz signal propagating vertically through your IRI ionosphere from Problem 1. Use your MSIS atmosphere profile from the previous Module to determine the collision frequency, and assume $|B_0| = 50,000$ nT.
9. **(10) ICON Mission:** The ICON Mission has four instruments: MIGHTI, IVM, EUV, and FUV (thank Berkeley for those creative acronyms). Do some research and pick your favorite instrument. In a couple of paragraphs, explain:
 - What does this instrument do?
 - What is the instrument SWAP (size, weight, and power)?
 - In basic terms, how does it work?
 - How does it contribute to the science goals of the ICON mission – in particular, what information about the ionosphere does it provide?